Thesis Project Summary

Unsupervised Clustering of Alzheimer's/Dementia Patients from Brain MRI

Goal

Using the OASIS cross-sectional dataset (416 patients, 5 MRI scan types per patient), the question is whether **image-derived features alone, or combined with clinical variables, produce patient clusters that align with clinical dementia severity (CDR)** — without ever using CDR as a clustering input. CDR (0 = healthy, 0.5 = very mild, 1 = mild, 2 = moderate) is known for only 235 of the 416 patients and is used strictly for **post-hoc evaluation** of whatever clusters come out.

Feature extraction

Each patient has five MRI planes (coronal, atlas-registered sagittal and transverse, a brain-masked transverse, and a native/unregistered sagittal). Each image is padded, resized, and converted to HOG features (9 orientations, 8×8 cells, 2×2 blocks → 8,100 features/image, 40,500 total per patient). Per image type, PCA is applied with a two-pass strategy that finds the minimum component count needed for ∼59% variance in the hardest type, then refits every type to that same number of components (110), so all five image types are directly comparable dimensionally.

Clustering framework

Clustering uses a family of robust, mixed-type distance metrics and a fast k-medoids engine (`robust_mixed_dist` / `db_robust_clust`, developed by our advisor Fabio Scielzo-Ortiz). Here it clusters PCA-reduced HOG features extracted from MRI images. The core methods:

- **Robust Mahalanobis** distance for single quantitative blocks
- **Generalised Gower** distance for mixing quantitative + binary + categorical clinical variables
- **Related Metric Scaling (RelMS)** for combining several *typed blocks* (e.g. five separate image types), each rescaled by its own geometric variability before being combined, so no single block dominates purely because it has more columns
- **SampleDistClustering / FasterPAM k-medoids**, which subsamples for tractability and returns real patients as cluster medoids (not synthetic centroids)

What was actually run

1. **Per-image-type clustering** (Euclidean and Robust Mahalanobis, $k = 3/4$, each of the 5 image types alone) — best single-type result was `masked_tra` (Euclidean, $k = 3$): ARI = 0.060, NMI = 0.085, the strongest CDR alignment found anywhere in the project.

2. **Merged 5-image clustering** (all 550 PCA features concatenated, Euclidean vs. Robust Mahalanobis, $k = 3/4$) — ARI stayed in $[-0.007, 0.013]$, NMI in $[0.02, 0.05]$.

3. **Clinical + imaging via Generalised Gower** (all-images-merged, each single image type, and clinical-only, all combined with 8 clinical variables) — revealed an anomaly: every one of

these variants (all-images+clinical, cor+clinical, gfc_sag+clinical, ..., and even the clinical-only baseline) produced numerically **identical** results (silhouette ≈ 0.328, ARI $= 0.016$, NMI $= 0.022$ at $k = 3$), down to the same cluster sizes. This strongly suggests the clinical block was dominating the Generalised Gower distance so completely that the 110–550 image PCA columns contributed essentially zero signal.

4. **Two-image concatenation** (masked_tra + cor, Euclidean) — ARI still near zero.

5. **RelMS distance** (Related Metric Scaling), introduced specifically to fix the block-dominance problem from step 3, applied to: all 5 images merged; all 5 images + clinical (complete cases, $n = 216$); each single image + clinical; and the two most promising two-image combinations (`masked_tra+cor`, `masked_tra+gfc_tra`), with and without clinical data. Best result: `masked_tra+cor+clinical`, $k = 4 \rightarrow$ ARI $= 0.041$, NMI $= 0.041$ — still weak.

6. **Image-type redundancy analysis** — correlated the pairwise-patient-distance matrices of the five image types against each other. Most redundant pair: `cor` $\leftrightarrow$ `gfc_sag` ($r = 0.60$). Most complementary/independent pair: `masked_tra` $\leftrightarrow$ `sbj_sag` ($r = 0.016$) — meaning the native (unregistered) sagittal images carry the most unique structural signal, distinct from everything else.

7. **Leave-one-image-out sweep** — systematically removed each image type from the RelMS 5-image blend to see which types help vs. hurt CDR alignment. Removing `masked_tra` or `sbj_sag` tended to *improve* ARI/NMI slightly (they add noise more than signal in the 5-way blend), while removing `gfc_sag` or `gfc_tra` tended to hurt it — but all deltas were small (≤ 0.01), i.e. within noise.

8. **Full cluster profiling** (5 images + clinical, complete cases $n = 216$) — beyond ARI/NMI, this notebook profiled each cluster's mean age, MMSE, brain volume (eTIV/nWBV), education, SES, and CDR/sex breakdown, with visualizations (boxplots, stacked CDR bars, a z-scored cluster-mean heatmap, 2D/3D PCA scatter, and effect-size (η^2/Cramér's V) rankings of which clinical features actually separate the clusters). It also surfaced a second anomaly: at $k = 3/4$ on a related Tucker configuration, Robust Mahalanobis produced degenerate singleton-outlier clusters (e.g. sizes $\{1, 414, 1\}$) with a deceptively high silhouette (~ 0.78) — an artifact of isolating 1–2 outliers, not real structure.

9. **Per-block distance sensitivity (newest addition)** — following an update from the advisor (`robust-mixed-dist` 0.1.27) allowing a *different* distance per quantitative block within RelMS, this was tested on the same 5-image + clinical setup: keeping the five image blocks on Euclidean while switching only the clinical block to Robust Mahalanobis, switching *all* quantitative blocks to Robust Mahalanobis, and switching only `masked_tra`. All four configurations were also compared against four alternative distance *functions* (Minkowski, Canberra, Pearson, plain Mahalanobis). Result: the distance choice does meaningfully change *who ends up in which cluster* (pairwise ARI between distance variants ranged from 0.11 to 0.98), but CDR alignment never moves outside the noise band already seen everywhere else (best ARI vs. CDR ≈ 0.03, best NMI ≈ 0.02) — so the distance metric was not the bottleneck.

Central finding

Across every method tried — single image type, merged images, Generalised Gower, RelMS, two-image combinations, leave-one-out sweeps, and multiple distance-function variants — **clustering does not meaningfully recover CDR groups**. ARI stays roughly in $[-0.02, +0.06]$ and NMI in $[0.01, 0.10]$ throughout, i.e. barely above what random cluster assignment would produce. The best-performing single result across the whole project was `masked_tra` alone with Euclidean distance at $k = 3$ (ARI $= 0.060$). This is a genuine negative result, not a bug: the same features that separate patients geometrically (image texture/shape via HOG+PCA) simply don't line up with the axis clinicians use (CDR severity), and swapping distance metrics, block weightings, or image combinations doesn't change that.

Two things worth flagging as anomalies (not fully resolved)

1. In the Generalised Gower experiments (step 3), all image+clinical variants were numerically identical to the clinical-only baseline — pointing to a likely scaling issue where the clinical block dominates that particular distance formulation.

2. In one Tucker-decomposition configuration, Robust Mahalanobis produced degenerate singleton clusters with an artificially high silhouette score — a reminder that silhouette alone can be misleading when a method isolates a couple of extreme outliers rather than finding real structure.